

Project on Hamiltonian Simulation - Part I

Exercise 1: Matrix Exponentials

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Exercise 1.1: Exponential of a diagonal matrix

Let $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ be an $n \times n$ diagonal matrix.

Starting with the definition of the matrix exponential:

$$e^D = \sum_{k=0}^{\infty} \frac{D^k}{k!} = I + D + \frac{D^2}{2!} + \frac{D^3}{3!} + \dots \quad (1)$$

For a diagonal matrix $D = \text{diag}(\lambda_1, \dots, \lambda_n)$, note that:

$$D^k = \text{diag}(\lambda_1^k, \lambda_2^k, \dots, \lambda_n^k) \quad (2)$$

for any $k \geq 0$.

Therefore:

$$\frac{D^k}{k!} = \text{diag}\left(\frac{\lambda_1^k}{k!}, \frac{\lambda_2^k}{k!}, \dots, \frac{\lambda_n^k}{k!}\right) \quad (3)$$

The sum can be distributed inside each diagonal entry:

$$e^D = \sum_{k=0}^{\infty} \frac{D^k}{k!} = \sum_{k=0}^{\infty} \text{diag}\left(\frac{\lambda_1^k}{k!}, \frac{\lambda_2^k}{k!}, \dots, \frac{\lambda_n^k}{k!}\right) \quad (4)$$

$$= \text{diag}\left(\sum_{k=0}^{\infty} \frac{\lambda_1^k}{k!}, \sum_{k=0}^{\infty} \frac{\lambda_2^k}{k!}, \dots, \sum_{k=0}^{\infty} \frac{\lambda_n^k}{k!}\right) \quad (5)$$

By definition of the scalar exponential, each diagonal entry (i, i) equals e^{λ_i} :

$$e^D = \text{diag}\left(e^{\lambda_1}, e^{\lambda_2}, \dots, e^{\lambda_n}\right) \quad (6)$$

Exercise 1.2: Transformation of exponential under basis change

Let $M = BAB^{-1}$ for $A, B \in \mathbb{C}^{n \times n}$, with B invertible.

Proof of the hint: First, showing that $M^k = BA^k B^{-1}$ for all $k \geq 0$.

For $M = BAB^{-1}$:

$$M^k = (BAB^{-1})(BAB^{-1}) \cdots (BAB^{-1}) \quad (k \text{ times}) \quad (7)$$

$$= BA(B^{-1}B)A(B^{-1}B)A \cdots AB^{-1} \quad (8)$$

$$= BAA \cdots AB^{-1} = BA^k B^{-1} \quad (9)$$

The intermediate $B^{-1}B$ terms vanish, leaving only the A matrices.

Main result: Applying the defining formula to M :

$$e^M = \sum_{k=0}^{\infty} \frac{M^k}{k!} = \sum_{k=0}^{\infty} \frac{BA^k B^{-1}}{k!} \quad (10)$$

Factoring B on the left and B^{-1} on the right from the sum:

$$e^M = B \left(\sum_{k=0}^{\infty} \frac{A^k}{k!} \right) B^{-1} \quad (11)$$

$$= B \left(I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \cdots \right) B^{-1} \quad (12)$$

The middle sum is simply e^A by definition:

$$e^M = B e^A B^{-1} \quad (13)$$

Exercise 1.3: Matrix exponential of tensor product with identity

Showing that for any square matrix A and the identity matrix I :

$$e^{A \otimes I} = e^A \otimes I \quad (14)$$

Starting with the definition:

$$e^{A \otimes I} = \sum_{k=0}^{\infty} \frac{(A \otimes I)^k}{k!} \quad (15)$$

Note that:

$$(A \otimes I)^k = (A \otimes I)(A \otimes I) \cdots (A \otimes I) \quad (k \text{ times}) = A^k \otimes I^k = A^k \otimes I \quad (16)$$

since $I^k = I$ for all k .

Therefore:

$$e^{A \otimes I} = \sum_{k=0}^{\infty} \frac{(A \otimes I)^k}{k!} = \sum_{k=0}^{\infty} \frac{A^k \otimes I}{k!} \quad (17)$$

$$= \left(\sum_{k=0}^{\infty} \frac{A^k}{k!} \right) \otimes I \quad (18)$$

The sum is e^A by definition:

$$e^{A \otimes I} = e^A \otimes I \quad (19)$$

Exercise 1.4: Counterexample for tensor products

Showing that in general, the relation $e^{A \otimes B} = e^A \otimes e^B$ does not hold.

Let:

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \quad (20)$$

Since both are diagonal matrices, using Exercise 1.1:

$$e^A = \begin{pmatrix} e & 0 \\ 0 & 1 \end{pmatrix}, \quad e^B = \begin{pmatrix} 1 & 0 \\ 0 & e \end{pmatrix} \quad (21)$$

By the definition of the Kronecker product:

$$e^A \otimes e^B = \begin{pmatrix} e & 0 & 0 & 0 \\ 0 & e^2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & e \end{pmatrix} \quad (22)$$

Now consider:

$$A \otimes B = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (23)$$

This matrix squares to itself: $(A \otimes B)^2 = A \otimes B$, and more generally $(A \otimes B)^k = A \otimes B$ for all $k \geq 1$.

Therefore:

$$e^{A \otimes B} = \sum_{k=0}^{\infty} \frac{(A \otimes B)^k}{k!} = I + \sum_{k=1}^{\infty} \frac{A \otimes B}{k!} \quad (24)$$

$$= I + (A \otimes B) \sum_{k=1}^{\infty} \frac{1}{k!} = I + (A \otimes B)(e - 1) \quad (25)$$

$$= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & e & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (26)$$

Clearly:

$$e^A \otimes e^B = \begin{pmatrix} e & 0 & 0 & 0 \\ 0 & e^2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & e \end{pmatrix} \neq \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & e & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = e^{A \otimes B} \quad (27)$$

This provides a counterexample showing that $e^{A \otimes B} \neq e^A \otimes e^B$ in general.

Exercise 1.5: Unitarity of quantum time evolution

Proving that for Hermitian $H \in \mathbb{C}^{n \times n}$ and $t \in \mathbb{R}_{\geq 0}$, the matrix exponential e^{-iHt} is unitary.

A matrix U is unitary when $UU^\dagger = U^\dagger U = I$.

Starting with the definition:

$$e^{-iHt} = \sum_{k=0}^{\infty} \frac{(-iHt)^k}{k!} \quad (28)$$

For the k -th term, note that $(-iHt)^k = (-i)^k H^k t^k$.

Taking the conjugate transpose:

$$\left[(-iHt)^k\right]^\dagger = \left[(-i)^k H^k t^k\right]^\dagger = i^k (H^k)^\dagger t^k = i^k H^k t^k \quad (29)$$

where $t \in \mathbb{R}$ (so t^k is real) and H is Hermitian (so $(H^k)^\dagger = H^k$).

Therefore:

$$\left[e^{-iHt}\right]^\dagger = \left[\sum_{k=0}^{\infty} \frac{(-iHt)^k}{k!}\right]^\dagger = \sum_{k=0}^{\infty} \frac{i^k H^k t^k}{k!} = e^{iHt} \quad (30)$$

Computing the product:

$$e^{-iHt} \cdot e^{iHt} = \left(\sum_{k=0}^{\infty} \frac{(-iHt)^k}{k!}\right) \left(\sum_{j=0}^{\infty} \frac{(iHt)^j}{j!}\right) \quad (31)$$

$$= \sum_{k=0}^{\infty} \sum_{j=0}^{\infty} \frac{(-i)^k i^j H^{k+j} t^{k+j}}{k! j!} \quad (32)$$

$$= \sum_{k=0}^{\infty} \sum_{j=0}^{\infty} \frac{(-1)^k i^{k+j} H^{k+j} t^{k+j}}{k! j!} \quad (33)$$

Let $n = k + j$. For fixed n , k ranges from 0 to n and $j = n - k$:

$$e^{-iHt} \cdot e^{iHt} = \sum_{n=0}^{\infty} \sum_{k=0}^n \frac{(-1)^k i^n H^n t^n}{k! (n-k)!} \quad (34)$$

$$= \sum_{n=0}^{\infty} \frac{(iHt)^n}{n!} \sum_{k=0}^n \binom{n}{k} (-1)^k \quad (35)$$

Using the binomial identity $(1-1)^n = \sum_{k=0}^n \binom{n}{k} (-1)^k = 0$ for $n > 0$ and equals 1 for $n = 0$:

$$e^{-iHt} \cdot e^{iHt} = \frac{(iHt)^0}{0!} \cdot 1 = I \quad (36)$$

Since $[e^{-iHt}]^\dagger = e^{iHt}$, this shows:

$$e^{-iHt} [e^{-iHt}]^\dagger = I \quad (37)$$

By symmetry, $[e^{-iHt}]^\dagger e^{-iHt} = I$ as well.

Therefore, e^{-iHt} is unitary.

Exercise 1.6: Circuit implementations

Part (a): Implementing $e^{-i\theta(Z \otimes Z)}$

The hint suggests thinking about how CNOT transforms the $Z \otimes Z$ operator. Computing the conjugation $\text{CNOT} \cdot (Z \otimes Z) \cdot \text{CNOT}$:

First, compute $\text{CNOT} \cdot (Z \otimes Z)$:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix} \quad (38)$$

Then multiply by CNOT on the right:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} = I \otimes Z \quad (39)$$

Therefore:

$$\text{CNOT} \cdot (Z \otimes Z) \cdot \text{CNOT} = I \otimes Z$$

Since $\text{CNOT}^{-1} = \text{CNOT}$ (CNOT is self-inverse), rearranging gives:

$$Z \otimes Z = \text{CNOT} \cdot (I \otimes Z) \cdot \text{CNOT}$$

This is exactly the form $M = BAB^{-1}$ from Exercise 1.2, where $M = Z \otimes Z$, $B = \text{CNOT}$, and $A = I \otimes Z$.

Multiplying both sides by the scalar $-i\theta$:

$$-i\theta(Z \otimes Z) = -i\theta \cdot \text{CNOT} \cdot (I \otimes Z) \cdot \text{CNOT} \quad (40)$$

Since $-i\theta$ is a scalar, it commutes with CNOT:

$$-i\theta(Z \otimes Z) = \text{CNOT} \cdot (-i\theta(I \otimes Z)) \cdot \text{CNOT} \quad (41)$$

By Exercise 1.2:

$$e^{-i\theta(Z \otimes Z)} = \text{CNOT} \cdot e^{-i\theta(I \otimes Z)} \cdot \text{CNOT} \quad (42)$$

By Exercise 1.3:

$$e^{-i\theta(I \otimes Z)} = e^{I \otimes (-i\theta Z)} = I \otimes e^{-i\theta Z} \quad (43)$$

By the definition of the rotation gate $R_Z(\phi) = e^{-i(\phi/2)Z}$:

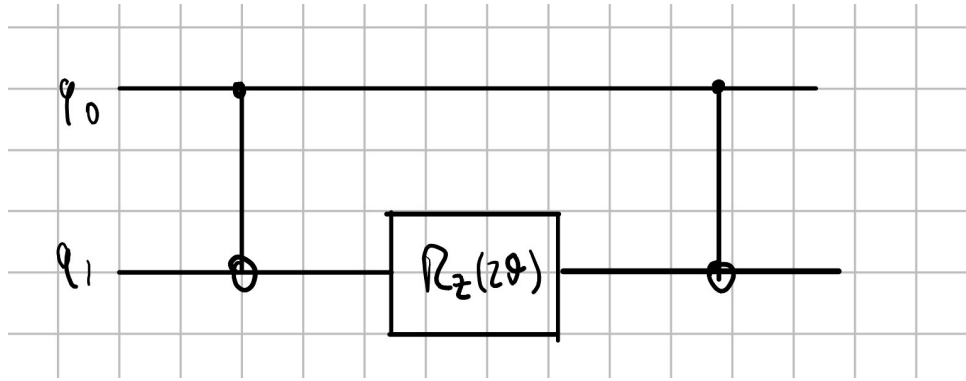
$$e^{-i\theta Z} = e^{-i(2\theta/2)Z} = R_Z(2\theta) \quad (44)$$

Therefore:

$$e^{-i\theta(Z \otimes Z)} = \text{CNOT} \cdot (I \otimes R_Z(2\theta)) \cdot \text{CNOT} \quad (45)$$

This shows that $e^{-i\theta(Z \otimes Z)}$ can be implemented using two CNOT gates (with q0 as control, q1 as target) and one $R_Z(2\theta)$ gate on q1.

Circuit diagram:



Part (b): Implementing $e^{-i\theta(X \otimes X)}$

Given that $HZH = X$, where H is the Hadamard gate, applying $H \otimes H$ on both sides of $Z \otimes Z$:

$$(H \otimes H)(Z \otimes Z)(H \otimes H) = (HZH) \otimes (HZH) = X \otimes X \quad (46)$$

Multiplying both sides by $-i\theta$:

$$-i\theta(X \otimes X) = (H \otimes H) \cdot (-i\theta(Z \otimes Z)) \cdot (H \otimes H) \quad (47)$$

By Exercise 1.2:

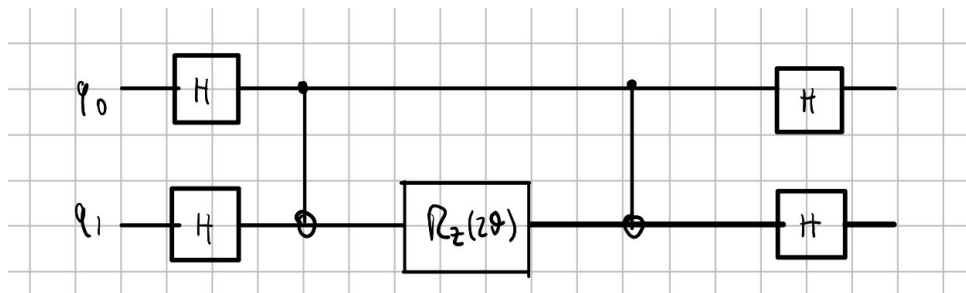
$$e^{-i\theta(X \otimes X)} = (H \otimes H) \cdot e^{-i\theta(Z \otimes Z)} \cdot (H \otimes H) \quad (48)$$

Substituting the result from part (a):

$$e^{-i\theta(X \otimes X)} = (H \otimes H) \cdot \text{CNOT} \cdot (I \otimes R_Z(2\theta)) \cdot \text{CNOT} \cdot (H \otimes H) \quad (49)$$

This circuit applies Hadamard gates to both qubits at the beginning and end, wrapping the circuit from part (a).

Circuit diagram:



Exercise 2.1: Show counterexample for matrix exponentials relation

We show that the relation

$$e^A e^B = e^{A+B}$$

does not always hold when A and B do not commute.

We consider the matrices:

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

First, let's compute AB and BA :

$$AB = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad BA = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Since

$$AB \neq BA,$$

the matrices do not commute.

Now, we can compute the exponentials.

Since $A^2 = 0$ and $B^2 = 0$, the exponential series simplify to

$$e^A = I + A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

and

$$e^B = I + B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

Their product is

$$e^A e^B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

Next, we compute $A + B$:

$$A + B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} =: X.$$

We compute powers of X :

$$X^2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I, \quad X^3 = X, \quad X^4 = I.$$

To make the calculation easier, we can split the exponential series into two different sums of the even and odd powers of X :

$$e^X = \sum_{k=0}^{\infty} \frac{X^k}{k!} = \sum_{k=0}^{\infty} \frac{X^{2k}}{(2k)!} + \sum_{k=0}^{\infty} \frac{X^{2k+1}}{(2k+1)!}.$$

Since $X^{2k} = I$ and $X^{2k+1} = X$, this becomes

$$e^X = \left(\sum_{k=0}^{\infty} \frac{1}{(2k)!} \right) I + \left(\sum_{k=0}^{\infty} \frac{1}{(2k+1)!} \right) X.$$

Here we recognize the series for hyperbolic functions,

$$\sum_{k=0}^{\infty} \frac{1}{(2k)!} = \cosh(1), \quad \sum_{k=0}^{\infty} \frac{1}{(2k+1)!} = \sinh(1),$$

and we obtain

$$e^{A+B} = \cosh(1) I + \sinh(1) X.$$

Therefore,

$$e^{A+B} = \cosh(1) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \sinh(1) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} \cosh(1) & \sinh(1) \\ \sinh(1) & \cosh(1) \end{pmatrix}.$$

Numerically,

$$\cosh(1) \approx 1.543, \quad \sinh(1) \approx 1.175,$$

so

$$e^{A+B} \approx \begin{pmatrix} 1.543 & 1.175 \\ 1.175 & 1.543 \end{pmatrix}.$$

Clearly, we can see by observation that

$$e^A e^B \neq e^{A+B}.$$

Therefore, the relation

$$e^A e^B = e^{A+B}$$

is not true in general when A and B do not commute.

Exercise 2.2: Bounding the error of “splitting off” a single H_j term

In this exercise, we are interested in the Trotterized Hamiltonian Simulation and we first want to prove the following claim:

$$D\left(e^{-iH\delta}, e^{-iH_1\delta} e^{-i\sum_{j=2}^m H_j\delta}\right) = \mathcal{O}\left(\delta^2 m h_{\max}^2\right).$$

To do so, we use the first-order Trotter formula:

$$D\left(e^{x(A+B)}, e^{xA} e^{xB}\right) = \mathcal{O}\left(x^2 \|A\|_{\text{op}} \cdot \|B\|_{\text{op}}\right).$$

We first rewrite the Hamiltonian H as

$$H = H_1 + \sum_{j=2}^m H_j.$$

Then, we can plug it back into the exponential, where we obtain

$$e^{-iH\delta} = e^{-i(H_1 + \sum_{j=2}^m H_j)\delta}.$$

Applying the Trotter formula with

$$A = -iH_1, \quad B = -i\sum_{j=2}^m H_j, \quad x = \delta,$$

gives us

$$D\left(e^{-iH\delta}, e^{-iH_1\delta} e^{-i\sum_{j=2}^m H_j\delta}\right) = \mathcal{O}\left(\delta^2 \|H_1\|_{\text{op}} \left\| \sum_{j=2}^m H_j \right\|_{\text{op}}\right).$$

And now, using the triangle inequality for the operator norm,

$$\left\| \sum_{j=2}^m H_j \right\|_{\text{op}} \leq \sum_{j=2}^m \|H_j\|_{\text{op}}.$$

Since

$$h_{\max} := \max_j \|H_j\|_{\text{op}},$$

we obtain

$$\|H_1\|_{\text{op}} \leq h_{\max}$$

and

$$\sum_{j=2}^m \|H_j\|_{\text{op}} \leq (m-1)h_{\max}.$$

Therefore,

$$D\left(e^{-iH\delta}, e^{-iH_1\delta} e^{-i\sum_{j=2}^m H_j\delta}\right) = \mathcal{O}\left(\delta^2 h_{\max}(m-1)h_{\max}\right).$$

Finally, simplifying the expression gives us

$$D\left(e^{-iH\delta}, e^{-iH_1\delta} e^{-i\sum_{j=2}^m H_j\delta}\right) = \mathcal{O}\left(\delta^2 m h_{\max}^2\right).$$

Hence, we have shown the desired result.

Exercise 2.3: Bounding the error of combining time slices

From the previous exercise, we can iteratively split off each exponential term and obtain the following bound:

$$D\left(e^{-iH\delta}, \prod_{j=1}^m e^{-iH_j\delta}\right) = \mathcal{O}\left(\delta^2 m^2 h_{\max}^2\right).$$

We now want to bound the error over a total simulation time t . To do so, we can divide the evolution into r equal time slices of size

$$\delta = \frac{t}{r}.$$

Thus, the exponential becomes

$$e^{-iHt} = \left(e^{-iH\delta}\right)^r.$$

Similarly, the Trotter approximation becomes

$$\left(\prod_{j=1}^m e^{-iH_j\delta}\right)^r.$$

We now use the subadditivity property of D , which implies that errors accumulate at most linearly when composing operators:

$$D(U_1 U_2, V_1 V_2) \leq D(U_1, V_1) + D(U_2, V_2).$$

Applying this to r successive time slices shows that the total error is bounded by the sum of the errors of each slice, hence it is at most r times the error of one slice.

$$D \left(e^{-iHt}, \left(\prod_{j=1}^m e^{-iH_j \delta} \right)^r \right) = \mathcal{O} \left(r \delta^2 m^2 h_{\max}^2 \right).$$

Substituting

$$\delta = \frac{t}{r},$$

back into our distance expression, we obtain the following:

$$D \left(e^{-iHt}, \left(\prod_{j=1}^m e^{-iH_j t/r} \right)^r \right) = \mathcal{O} \left(r \left(\frac{t}{r} \right)^2 m^2 h_{\max}^2 \right).$$

Finally, simplifying the expression gives us

$$D \left(e^{-iHt}, \left(\prod_{j=1}^m e^{-iH_j t/r} \right)^r \right) = \mathcal{O} \left(\frac{t^2}{r} m^2 h_{\max}^2 \right).$$

Hence, we conclude that

$$D \left(e^{-iHt}, \left(\prod_{j=1}^m e^{-iH_j t/r} \right)^r \right) = \mathcal{O} \left(\frac{t^2}{r} m^2 h_{\max}^2 \right).$$